

FORESTS

The lungs of our planet

Roughly one-third of the world's land area is covered with trees. Some are the largest and longest-living organisms on Earth. Their roots, trunks, branches, and leaves form an uncountable variety of microhabitats, the character of which varies according to location. Dead and decaying leaves and wood also form a vital component of forest ecosystems, providing habitat and food, and releasing nutrients back into forest soils. Clearings left by fallen trees throng with light-loving ground plants and insects until new trees close the gap.

Boreal and temperate forests

The wide range of climates in temperate areas supports coniferous, deciduous, broadleaf evergreen, and mixed forest. In the far northern boreal forests, winters are longer, temperatures lower, and snow fall more frequent—conditions to which coniferous trees are well adapted. Their triangular shape and narrow leaves prevent excess snow settling on their branches and

breaking them. By keeping their dark green leaves all year, they can make food whenever the Sun shines. The resin-filled leaves are distasteful to all but a few insects and so are not eaten even when food is scarce.

Farther south, winters are still cold, but summers are longer and warmer. Forests here are generally deciduous. The trees have broad leaves and spreading branches that maximize their ability to harvest light and get



BOREAL

The hardy, evergreen conifers found in northern boreal forests provide less food than other trees due to the harsh climate and short growing season. In winter, when food is scarce, many animals migrate to warmer areas or hibernate.



TEMPERATE CONIFEROUS

Nonflowering plants, such as conifers, produce their seeds in cones, which are released when dry conditions cause the cones to open. The tiny seeds of western hemlock are eaten by chickadees, pine siskins, and deer mice.



TEMPERATE BROADLEAF

The seasonal availability of some foods presents a challenge to woodland animals. Some, such as grey squirrels, solve the problem by hoarding nuts and seeds in tree holes and underground caches, to which they return in winter.

SEYCHELLES ▸

Tropical rainforest on Silhouette Island, Seychelles, in the Indian Ocean.

